Guidelines for Welding Notes & Symbols
Basic Training Material

1. Introduction
2. Checking correctness of weldment model for welding requirements
3. Types of welding and types of welded joints preferred at PARI
4. Understanding classification of part and its relevance with category of parts at PARI
5. Deciding Welding quality requirements
6. General Welding Details for unspecified Weld joints
7. Welding notes on part drawing
8. Welding Symbols on drawing
9. Details for “on assembly welding”
10. Details for M.S. to Alloy steel welding
11. Correctness & Completeness Review

1. Introduction

These guidelines applies for

- Steel Fabricated parts (Manufactured)
- On Assembly welding

These guidelines not applicable for

- Casting
- Non Ferrous
- Non metals

2. Correctness of Weldment model

- Check weldment model for welding requirements such as, stitch welding, full welding and leak proof welding.
- Check weldment part’s weld bead size.
- Check weldment model for any welding quality requirement. Refer slide no. 4 for welding quality levels, symbols, specifications and other details.
- Check weld inspection requirements such as Dye penetration, Ultrasonic, destructive, Weld Penetration test on test coupons of required size etc. on weldment model.
- Check weldment model materials for all parts. Generally materials should be homogeneous either in m.s. or alloy steels; But we can also weld.
- Review of any specific welding notes provided in keywords / comments on model.
- Check Stress relieving with type, Thermal /vibratory and shot Blasting requirements (For Heavy Structures / T+S parts, Functionally critical parts) are provided in keywords / comments on model.
- Weld Details are mandatory from AK2 for T+S Category parts.

3a. Types of Welding generally used at PARI

1. SMAW - Shielded Metal Arc Welding
2. CO2 MIG – CO2 Metal Inert Gas Welding
3. RSW – Resistive Spot Welding
4. TIG - Tungsten Inert Gas Welding

3b. Types of Welding joints preferred at PARI

1. Fillet Weld



2. Butt Weld- There are many types of butt welds like



a) V Butt Welding



b) U Butt Welding



c) Bevel Welding etc.,

but all these types fall within one of categories: single welded butt joints, double welded butt joint

4. Understanding classification of part and its relevance with category of parts at PARI

Classification of Parts-

1) "T + S" Class parts = Are critical stressed parts in Tooling having safety issue

2) "A" Class parts = Are performance & safety critical parts i.e. these are the parts that affect the performance & safety of the equipment or final product.

3) "B" Class parts = Are function critical parts & affect assembly area i.e. these are parts that affect the function and assembly of the equipment or final product.

4) "C" Class parts = Are parts with all non-critical parameters & not affecting the function, performance & safety of the equipment or final product.

- For part class examples, refer "Guideline for deciding Classification of Parts PRD-GD-DES-03"

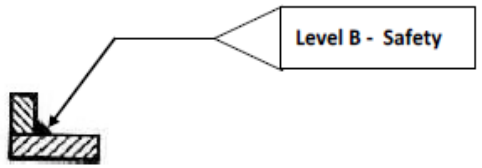
5. Deciding Welding Quality Requirements

Quality Levels for Weld Imperfections -

Level Symbol	Quality Level	Classification
B	Stringent	i) Applicable for Welded Joints in Safety Critical parts and Structures that affect the safety of the equipment/ final product e.g. Press, Straight column, X Beam, Y Beam etc. **OR** as per the Contractual requirements.
		ii) Applicable for Welded Joints that are critical to Human Safety. Designer **MUST** identify and define on the drawings any welds that are critical to human safety.
C	Intermediate	i) Applicable for Welded Joints in Functional /Performance Critical parts that affect the assembly and/or performance of the equipment /final product e.g. Loaded and movable structures, Base Frame, Machine table, Turn table etc. **OR** as per the Contractual requirements.
		ii) Applicable for Welded Joints in **"A"** and **"B"** Class Parts.
D	Moderate	i) Applicable for Welded Joints in Non-functional and Non-critical parts e.g. Sensor Bracket, Bracket for pallet, Fencing, covers etc.
		ii) Applicable for Welded Joints in **"C"** Class Parts.

On Fabrication drawings - Following weld symbol is required to be specified for **"B"** or **"C"** Quality Levels of Welded Joints, with the reason (e.g. Safety, Function etc).

For example -



- Provide Weld class as per table

Weld Class	Classification of parts
B	T+S
C	A & B
D	C

- For more details on weld class and its inspection, Refer ISO 5817.

6. General Welding Details for unspecified Weld joints.

1) Refer Table for Unspecified Bead Size.

UNSPECIFIED WELD BEAD SIZES IN mm		
MIN. PLATE / SECTION THICKNESS	BEAD SIZE FOR WELD CLASS C & D	BEAD SIZE FOR WELD CLASS B
UPTO 4	2	2
4 UPTO 8	3	4
8 UPTO 12	5	6
ABOVE 12	5	8

2) Refer table below for unspecified weld details.

Weld length (mm)	Weld Class C & D(weld-Pitch)	Weld class B (weld-Pitch)
Up to 250	Full length weld	Full length weld
250 to 500	50-100	50-100
500 to 1000	50-150	50-150
Above 1000	100-200	100-200

3) For stitch welding, Apply 25 mm weld on end of weld seam unless specified.

- These above details / tables are applicable when AK2 does not provide any welding details
- These General welding details shall not applicable for T+S category parts.

Contd…

7. Welding Notes on part drawing

A) "T + S" Class parts -

WELDING NOTES -
1) ALL JOINTS SHOULD BE AS PER WELD SYMBOLS MENTIONED ON DRAWING.
2) ALL CHILD PARTS TOLERANCE SHOULD BE AS PER IS-2102 COARSE GRADE UNLESS OTHERWISE SPECIFIED
3) ALL CHILD PARTS TO BE STRAIGHTENED BEFORE WELDING
4) SUITABLE EDGE PREPARATION TO BE DONE BEFORE WELDING
5) ALL PARTS TO BE TACK WELDED & CHECKED FOR DIMENSION BEFORE FULL WELDING
6) MAXIMUM DISTORTION (TWIST & BEND) ALLOWED OVER FULL LENGTH IS 3MM OR 01 MM / MTR WHICHEVER IS MINIMUM (AFTER FULL WELDING)
7) FABRICATED STRUCTURE TOLERANCES SHOULD BE AS PER IS 2102 COARSE GRADE UNLESS SPECIFIED.
8) WELDING SHOULD BE NEAT & CLEAN
9) ALL WELDING JOINTS SHALL BE DRESSED PROPERLY
10) MASK MACHINED FACES (IF ANY) DURING PAINTING
11) WELD CLASS - "B"
12) STRESS RELIEVING - APPLICABLE
13) SHOT BLASTING - APPLICABLE / NOT APPLICABLE

INSPECTION NOTES -
1) DYE PENETRATION TEST FOR WELDING - APPLICABLE / NOT APPLICABLE
2) ULRASONIC TESTING FOR FINISH JOB - APPLICABLE / NOT APPLICABLE
3) LEAK PROOF WELD JOINT - REQUIRED / NOT REQUIRED

Contd...

7. Welding Notes on part drawing…..Contd

B) “A” Class parts -

WELDING NOTES -
1) ALL JOINTS SHOULD BE AS PER TABLES “GENERAL WELDING DETAILS”
2) ALL CHILD PARTS TOLERANCE SHOULD BE AS PER IS-2102 COARSE GRADE UNLESS OTHERWISE SPECIFIED
3) ALL CHILD PARTS TO BE STRAIGHTENED BEFORE WELDING
4) SUITABLE EDGE PREPARATION TO BE DONE BEFORE WELDING
5) ALL PARTS TO BE TACK WELDED & CHECKED FOR DIMENSION BEFORE FULL WELDING
6) MAXIMUM DISTORTION (TWIST & BEND) ALLOWED OVER FULL LENGTH IS 3MM OR 01 MM / MTR WHICHEVER IS MINIMUM (AFTER FULL WELDING)
7) FABRICATED STRUCTURE TOLERANCES SHOULD BE AS PER IS 2102 COARSE GRADE UNLESS SPECIFIED.
8) WELDING SHOULD BE NEAT & CLEAN
9) ALL WELDING JOINTS SHALL BE DRESSED PROPERLY
10) MASK MACHINED FACES (IF ANY) DURING PAINTING
11) WELD CLASS - "C"
12) STRESS RELIEVING - APPLICABLE / NOT APPLICABLE
13) SHOT BLASTING - APPLICABLE / NOT APPLICABLE

INSPECTION NOTES -
1) DYE PENETRATION TEST FOR WELDING - APPLICABLE / NOT APPLICABLE
2) ULRASONIC TESTING FOR FINISH JOB - APPLICABLE / NOT APPLICABLE
3) LEAK PROOF WELD JOINT - REQUIRED / NOT REQUIRED

Contd…

7. Welding Notes on part drawing….Contd

C) “B” Class parts -

WELDING NOTES -

1) ALL JOINTS SHOULD BE AS PER TABLES “GENERAL WELDING DETAILS”
2) ALL CHILD PARTS TO BE STRAIGHTENED BEFORE WELDING
3) SUITABLE EDGE PREPARATION TO BE DONE BEFORE WELDING
4) FABRICATED STRUCTURE TOLERANCES SHOULD BE AS PER IS 2102 COARSE GRADE UNLESS SPECIFIED.
5) WELDING SHOULD BE NEAT & CLEAN
6) ALL WELDING JOINTS SHALL BE DRESSED PROPERLY
7) MASK MACHINED FACES (IF ANY) DURING PAINTING
8) WELD CLASS - "C"
9) STRESS RELIEVING - APPLICABLE / NOT APPLICABLE
10) SHOT BLASTING - APPLICABLE / NOT APPLICABLE

INSPECTION NOTES -

1) DYE PENETRATION TEST FOR WELDING - APPLICABLE / NOT APPLICABLE
2) ULRASONIC TESTING FOR FINISH JOB - APPLICABLE / NOT APPLICABLE
3) LEAK PROOF WELD JOINT - REQUIRED / NOT REQUIRED

Contd…

7. Welding Notes on part drawing….Contd

D) “C” Class parts -

WELDING NOTES -
1) ALL JOINTS SHOULD BE AS PER TABLES “GENERAL WELDING DETAILS”
2) SUITABLE EDGE PREPARATION TO BE DONE BEFORE WELDING
3) FABRICATED STRUCTURE TOLERANCES SHOULD BE AS PER IS 2102 COARSE GRADE UNLESS SPECIFIED.
4) WELDING SHOULD BE NEAT & CLEAN
5) WELD CLASS - "D"

- In all welding notes, user has to choose only one details from the options below

1) Applicable / Not Applicable
2) Required / Not Required.

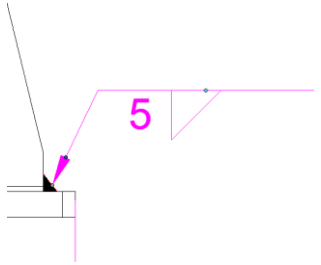
- To copy welding notes on your draft files, Refer file


Solid Edge Draft Document

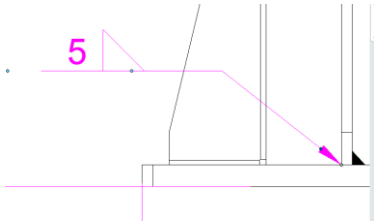
8. Welding Symbols on drawing

Welding Symbols

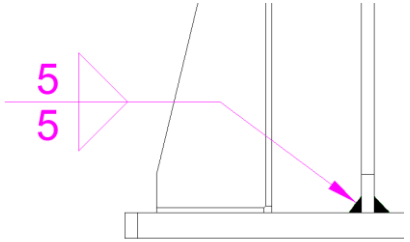
1) Fillet weld on arrow side with 5 mm size



2) Fillet weld on other side with 5 mm size



3) Fillet weld on both side with 5 mm size



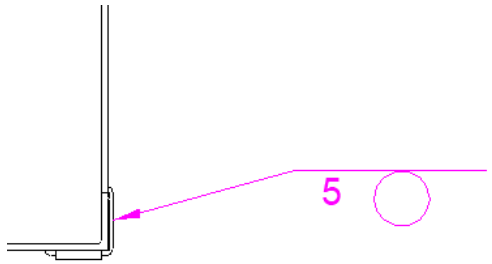
4) Convex contoured Fillet weld on arrow side with 5 mm size, 100 mm length and 200 mm pitch.



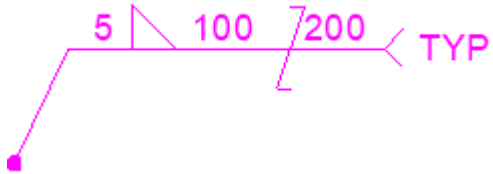
Contd…

8. Welding Symbols on drawing …..Contd

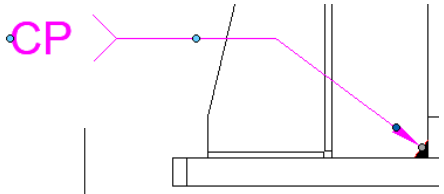
5) Spot weld on arrow side (Always) with 5 mm size



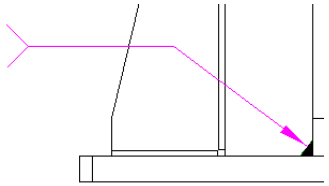
6) Fillet weld on other side with 5 mm size, 100 mm length and 200 mm Gap. And same welding is used at typical locations.



7) Welding with complete penetration



8) Process used for welding CO2-MIG



- Welding Symbols to be provided on drawing when AK2 has specified particular welding requirement for certain joint.

Contd…

8. Welding Symbols on drawing…..Contd

Important points-

- The reference line of the welding symbol is used to designate the type of weld to be made, its location, dimensions, extent, contour, and other supplementary information.
- Any welded joint indicated by a symbol will always have an arrow side and an other side. Accordingly, the terms arrow side, other side, and both sides are used herein to locate the weld with respect to the joint.
- The tail of the welding symbol is used for designating the welding and cutting processes as well as the welding specifications, procedures, or the supplementary information to be used in making the weld.
- If notations are not used, the tail of the symbol may be omitted.
- Similar to fillet weld we can use symbols for Butt Weld
- Like convex, flush & concave weld contours to be used on symbols as per requirement

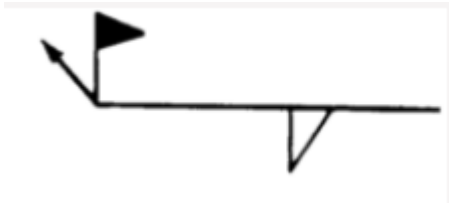
9. Details for “on assembly welding”

- On site / field welding- Check if any specific ‘on assembly welding’ is required due to assembly requirement, then that area need to be mention on assembly drawing by note indicating the area by arrow.

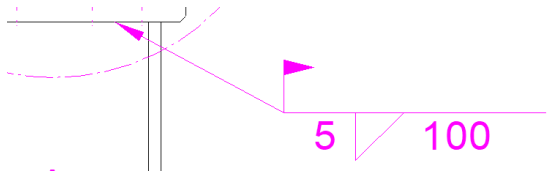
  On Assembly Welding Note-

  “Parts are to be welded as per General Welding Details during assembly and welding should be neat and clean”

- For on site Welding, welding Symbols also can be shown as specified in welding symbol below.



Example-



Joint shown by arrow to be on assembly welded with 5 mm weld size and 100 mm length.

10. Details for M.S. to Alloy steel welding

- For Steel to low alloy welding, Parts should be GMAW welded with welding rod no. 7018.
- For Steel to high alloy welding, Parts should be GMAW welded with welding rod no. 8018.

11. Correctness and completeness review

Checker to confirm

- All welding requirements are meeting and accordingly welding note and/or welding symbols are specified on drawing.
- BOM is correct for size, thickness, and section details.
- All child parts locating dimensions are provided.
- All welding details & notes are correctly chosen and provided.
- Generally machining should be on different sheet and IS tolerance grade to be mentioned on that Sheet. If not, machining dimensions are provided with required tolerances (Not kept open because we specified IS-2102 coarse grade for fabrication dimensions, which can be mistaken).